

Competitive Programming Tree & Graph Roadmap

Goal: Progress from beginner to Codeforces Expert by mastering tree and graph techniques in a structured order.

Phase 1: DFS/BFS Fundamentals

- LC 559 - Maximum Depth of N-ary Tree
- LC 690 - Employee Importance
- LC 1443 - Minimum Time to Collect All Apples in a Tree
- LC 785 - Is Graph Bipartite
- LC 841 - Keys and Rooms
- CSES: Counting Rooms, Building Roads, Message Route

Phase 2: Tree Height & Diameter

- LC 1245 - Tree Diameter
- LC 1522 - Diameter of N-ary Tree
- LC 2246 - Longest Path With Different Adjacent Characters
- LC 687 - Longest Univalue Path
- CSES: Tree Diameter

Phase 3: Subtree DP

- LC 1519 - Number of Nodes in the Subtree With the Same Label
- LC 2049 - Count Nodes With the Highest Score
- LC 3331 - Find Subtree Sizes After Changes
- CSES: Subordinates, Tree Distances I

Phase 4: Tree DP

- LC 2538 - Difference Between Maximum and Minimum Price Sum
- LC 2646 - Minimize the Total Price of Trips
- LC 2467 - Most Profitable Path in a Tree
- CSES: Tree Matching

Phase 5: Rerooting DP

- LC 834 - Sum of Distances in Tree
- LC 2581 - Count Number of Possible Root Nodes
- CSES: Tree Distances II

Phase 6: Binary Lifting / LCA

- CSES: Company Queries I
- Company Queries II
- Distance Queries
- CF EDU LCA

Phase 7: Euler Tour

- CSES: Subtree Queries
- Path Queries
- CF Euler Tour practice

Phase 8: DSU on Tree (Small-to-Large)

- CF 600E - Lomsat Gelral
- CF 246E - Blood Cousins Return

Phase 9: Heavy-Light Decomposition

- CSES: Path Queries II
- CF 343D - Water Tree
- QTREE (SPOJ)

Phase 10: Centroid Decomposition

- CF 342E - Xenia and Tree
- SPOJ QTREE5

Phase 11: Graph Algorithms

- Topological Sort
- SCC (Kosaraju & Tarjan)
- Bridges & Articulation Points
- Dijkstra
- 0-1 BFS
- Bellman-Ford
- Floyd-Warshall
- Minimum Spanning Tree
- Functional Graphs

Recommended LeetCode Set

559, 690, 1443, 1245, 1522, 1519, 2049, 2246, 834, 2581, 2538, 2646, 2467, 2440, 1617, 785, 802, 841, 797, 1557, 1857, 2328, 329, 2065, 1786.

Recommended External Resources

CSES Tree Algorithms (complete section)

Codeforces EDU: DSU, Trees, LCA

AtCoder Educational DP Contest (Tree DP problems)

SPOJ QTREE series for HLD.